

50-346

NRC DISTRIBUTION FOR PART 50 DOCKET MATERIAL

FILE NUMBER

TO:

Mr. John F. Stolz

FROM:

Toledo Edison Company
Toledo, Ohio
Lowell E. Roe

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4/6/77

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DESCRIPTION

Ltr. trans the following:

ACKNOWLEDGED

(1-P)

PLANT NAME:

Davis-Besse

RJL

DO NOT REMOVE

ENCLOSURE

Consists of information concerning
the Unit No. 1 Safety Analysis....

(5-P)

SAFETY

FOR ACTION/INFORMATION

ENVIRO

ASSIGNED AD:	Vassallo	ASSIGNED AD:
BRANCH CHIEF:	Stolz	BRANCH CHIEF:
PROJECT MANAGER:	Engle	PROJECT MANAGER:
LIC. ASST. :	Hylton	LIC. ASST. :

INTERNAL DISTRIBUTION

REG FILE	SYSTEMS SAFETY	PLANT SYSTEMS	SITE SAFETY &
NRC PDR	HEINEMAN	TEDESCO	ENVIRO ANALYSIS
I & E (2)	SCHROEDER	BENAROYA	DENTON & MULLER
OELD		LAINAS	
GOSSICK & STAFF	ENGINEERING	IPPOLITO	ENVIRO TECH.
MIPC	MACARRY	KIRKWOOD	ERNST
CASE	BOSNA		BALLARD
HANAUER	SIHWE L	OPERATING REACTORS	YOUNGBLOOD
HARLESS	PAWLICKI	STELLO	
			SITE TECH.
PROJECT MANAGEMENT	REACTOR SAFETY	OPERATING TECH.	GAMMILL (2)
BOYD	ROSS	EISENHUT	STAPP
P. COLLINS	NOVAK	SHAO	HULMAN
HOUSTON	ROSZTOCZY	BAER	
PETERSON	CHECK	BUTLER	SITE ANALYSIS
MELTZ		GRIMES	VOLLMER
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CONTROL NUMBER

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ACRS 16 CYS HOLDING/SENT	AS CAT A	

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Docket No. 50-346

April 6, 1977

Serial No. 261



LOWELL E. ROE

Vice President
Facilities Development
(419) 259-5242

Director of Nuclear Reactor Regulation
Attn: Mr. John F. Stolz, Chief
Light Water Reactor Branch No. 1
Division of Project Management
United States Nuclear Regulatory Commission
Washington, D.C. 20555

REGULATORY DOCKET FILE COPY

Dear Mr. Stolz:

The Davis-Besse Nuclear Power Station Unit 1 Technical Specification Bases 2.2.1 states in part that "...each Allowable Value is equal to or less than the drift allowance assumed for each trip in the Safety Analyses." In general the Safety Analyses took credit for periodic testing and calibration of the instrumentation. A conservative drift rate was applied to the trip setpoint along with other instrument uncertainties to determine the value that the parameter assumed in the Safety Analyses. This conservative drift rate would then give a different drift for a monthly Channel Functional Test than for an 18 month Channel Calibration. Therefore, the attached changes to the Davis-Besse Nuclear Power Station Unit 1 Technical Specifications provide these dual allowable values which will also be included in the next revision of the FSAR.

Yours very truly,

Attachment

dh c/18

~~77108004~~

TABLE 2.2-1

REACTOR PROTECTION SYSTEM INSTRUMENTATION TRIP SETPOINTS

<u>FUNCTIONAL UNIT</u>	<u>TRIP SETPOINT</u>	<u>ALLOWABLE VALUES</u>
1. Manual Reactor Trip	Not Applicable	Not Applicable
2. High Flux	$\leq 105.5\%$ of RATED THERMAL POWER with four pumps operating $\leq 80.7\%$ of RATED THERMAL POWER with three pumps operating $\leq 53.0\%$ of RATED THERMAL POWER with one pump operating in each loop	$\leq 105.6\%$ of RATED THERMAL POWER with four pumps operating * * * $\leq 80.8\%$ of RATED THERMAL POWER with three pumps operating * * * $\leq 53.1\%$ of RATED THERMAL POWER with one pump operating in each loop * * *
3. RC High Temperature	$\leq 619^{\circ}\text{F}$	≤ 619.08 * * *
4. Flux - Δ Flux-Flow ⁽¹⁾	Trip Setpoint not to exceed the limit line of Figure 2.2-1.	Allowable Values not to exceed the limit line of Figure 2.2-2. * * *
5. RC Low Pressure ⁽¹⁾	≥ 1985 psig	≥ 1976.5 psig * ≥ 1984.0 psig * *
6. RC High Pressure	≤ 2355 psig	≤ 2363.5 psig * ≤ 2356.0 * *
7. RC Pressure-Temperature ⁽¹⁾	$\geq (13.01 T_{\text{out}}^{\circ}\text{F} - 5973)$ psig	$\geq (13.01 T_{\text{out}}^{\circ}\text{F} - 5973.64)$ psig * * *

* For Channel Calibration

* * For Channel Functional Testing

* * * The allowable value for channel calibration is the same as the allowable value for channel functional testing.

TABLE 2.2-1

REACTOR PROTECTION SYSTEM INSTRUMENTATION TRIP SETPOINTS

<u>FUNCTIONAL UNIT</u>	<u>TRIP SETPOINT</u>	<u>ALLOWABLE VALUES</u>
8. High Flux/Number of Reactor Coolant Pumps On ⁽¹⁾	$\leq 55.0\%$ of RATED THERMAL POWER with one pump operating in each loop $\leq 0.0\%$ of RATED THERMAL POWER with two pump operating in one loop and no pumps operating in the other loop $\leq 0.0\%$ of RATED THERMAL POWER with no pumps operating or only one pump operating	$\leq 55.28\%$ of RATED THERMAL POWER with one pump operating in each loop * * * $\leq 0.28\%$ of RATED THERMAL POWER with two pumps operating in one loop and no pump operating in the other loop * * * $\leq 0.28\%$ of RATED THERMAL POWER with no pumps operating or only one pump operating * * *
9. Containment Pressure High	≤ 4 psig	≤ 4 psig * * *

(1) Trip may be manually bypassed when RCS pressure ≤ 1820 psig by actuating Shutdown Bypass provided that:

- The High Flux Trip Setpoint is $\leq 5\%$ of RATED THERMAL POWER
- The Shutdown Bypass High Pressure Trip Setpoint of ≤ 1820 psig is imposed, and
- The Shutdown Bypass is removed when RCS Pressure > 1820 psig.

* For Channel Calibration

* * For Channel Functional Testing

* * * The allowable value for channel calibration is the same as the allowable value for channel functional testing.

TABLE 3.3-4

SAFETY FEATURES ACTUATION SYSTEM INSTRUMENTATION TRIP SETPOINTS

<u>FUNCTIONAL UNIT</u>	<u>TRIP SETPOINT</u>	<u>ALLOWABLE VALUES</u>
INSTRUMENT STRINGS		
a. Containment Radiation	$\leq$ * mR/hr	$\leq$ * mR/hr * *
b. Containment Pressure - High	$\leq$ 18.4 psia	$\leq$ 18.52 psia * *
c. Containment Pressure - High-High	$\leq$ 38.4 psia	$\leq$ 38.52 psia * *
d. RCS Pressure - Low	$\geq$ 1620.75 psig	$\geq$ 1615.75 psig * *
e. RCS Pressure - Low-Low	$\geq$ 420.75 psig	$\geq$ 415.75 psig * *
f. BWST Level	$\geq$ 49.5 and $\leq$ 55.0 in. H ₂ O	$\geq$ 48.3 and $\leq$ 56.7 in. H ₂ O *
SEQUENCE LOGIC CHANNELS		
a. Essential Bus Feeder Breaker Trip (90%)	$\geq$ 3744 volts for 10 $\pm$ 1.5 sec	$\geq$ 3558 volts for for 10 $\pm$ 1.5 sec*
b. Diesel Generator Start, Load Shed on Essential Bus (59%)	$\geq$ 2071 and $\leq$ 2450 volts for 0.5 $\pm$ 0.1 sec	$\geq$ 2071 and $\leq$ 2450 volts for 0.5 $\pm$ 0.1 sec
INTERLOCK CHANNELS		
a. Decay Heat Isolation Valve	$\leq$ 530 $\geq$ 381 psig	$\leq$ 535 * * $\geq$ 376 * * psig

* To be determined after full power testing

* * The allowable value for channel calibration is the same as the allowable value for channel functional testing

TABLE 3.3-12

STEAM AND FEEDWATER RUPTURE CONTROL SYSTEM
INSTRUMENTATION TRIP SETPOINTS

<u>FUNCTIONAL UNITS</u>	<u>TRIP SETPOINTS</u>	<u>ALLOWABLE VALUES</u>
1. Steam Line Pressure - Low	≥ 591.6 psig	≥ 586.6 psig * * *
2. Steam Generator Level - Low*	$\geq 20"$ H ₂ O	≥ 591.6 psig * * * * $\geq 20"$ H ₂ O * * * * *
3. Steam Generator Feedwater Differential Pressure - High**	≥ 197.6 psid	≥ 199.6 psid * * * ≥ 197.6 psid * * * * *
4. Reactor Coolant Pumps - Loss of	High ≤ 1384.6 amps Low ≥ 106.5 amps	≤ 1384.6 amps * * * * * ≥ 106.5 amps * * * * *

*Measured above the lower steam generator tubesheet.

**Where differential pressure is steam generator minus feedwater pressure.

* * * For Channel Calibration

* * * * For Channel Functional Test

* * * * * The allowable value for channel calibration is the same as the allowable value for channel functional testing.